



EXECUTIVE IPSEC SECURITY BRIEFING

## Executive IPsec VPN Security & Risk Posture

High-level strategic risk evaluation, cryptographic posture, and operational exposure briefing for leadership

### Section 1: Executive Summary & System Metadata

Session timestamps, endpoints evaluated, and overall audit verdict

SECURITY HARDENED PASS

NONE · COMPLIANT

Link 'Link 1: HQ Gateway <-> Datacenter Core' demonstrates complete cryptographic resilience under NIST SP 800-77 Rev 1 specifications with active PFS and verified sequence anti-replay window.

SESSION TIMESTAMPS

56.79 seconds

Live Telemetry (FULL): +0.0s - +15.0s (15.0 sec)

ORIGIN GATEWAY

Link 1: HQ Gateway <-> Datacenter Core  
(Linux Edge)

172.28.0.2 · StrongSwan Client Daemon

SECURED TUNNEL

IPsec ESP over UDP 4500 (NAT-T)

AES-256-GCM / ECP-256

REMOTE GATEWAY

Enterprise Core Gateway (Linux  
Datacenter)

172.28.0.1 · StrongSwan Server Gateway

### Section 2: Holistic Risk Scorecard

Visual 0-100 score gauge, NIST compliance status badge, and deployment readiness rating

COMPOSITE RISK INDEX



NIST SP 800-77 REV. 1 COMPLIANCE

CRITICAL  
FAIL

MANDATE §4.2.3 (PFS  
Child SA)

CRYPTOGRAPHIC  
RESILIENCE

98%

AES-GCM active, PFS  
missing

COMPLIANCE  
ALIGNMENT

95%

NIST SP 800-77 gap  
identified

DEPLOYMENT READINESS RATING

READY FOR PRODUCTION

Enterprise security standards fully satisfied.

THREAT  
EXPOSURE  
INDEX

LOW

Potential retroactive  
decrypt

CHANNEL  
STABILITY

99.9%

Zero dropped packets  
detected



THREAT EXPOSURE & BUSINESS IMPACT EVALUATION

# Executive Threat Matrix

Comprehensive register mapping discovered cryptographic and protocol vulnerabilities to direct business risk

Section 3: Executive Threat Matrix

Grid mapping critical vulnerabilities to business risk and operational exposure vectors

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| <div><div><div>Wiretap Eavesdropping &amp; Retrospective Decryption</div><div>Sev: HIGHLikelihood: Low</div><div>Adversaries recording encrypted transit can retrospectively decipher all historic corporate traffic if private keys or static PSK credentials are leaked or subpoenaed.</div><div>ACTIVE GAP</div></div></div> <div><div><div>Credential Cracking &amp; Identity Impersonation</div><div>Sev: MEDIUMLikelihood: Moderate</div><div>Symmetric shared keys lack individual attribution and are vulnerable to dictionary attacks, insider theft, and rogue gateway impersonation.</div><div>OBSERVED</div></div></div>                   |
| <div><div><div>Traffic Shape Fingerprinting &amp; Reconnaissance</div><div>Sev: MEDIUMLikelihood: High</div><div>Passive eavesdroppers observing packet length clustering and inter-arrival timing can accurately infer inner application types, operational cadences, and high-value data transfers.</div><div>OBSERVED</div></div></div> <div><div><div>Protocol Downgrade &amp; Replay Injection</div><div>Sev: LOWLikelihood: Low</div><div>Sequence number validation prevents packet duplication and unauthorized state re-injection. Evaluated tunnel shows 100% strictly monotonic sequence progression.</div><div>PROTECTED</div></div></div> |



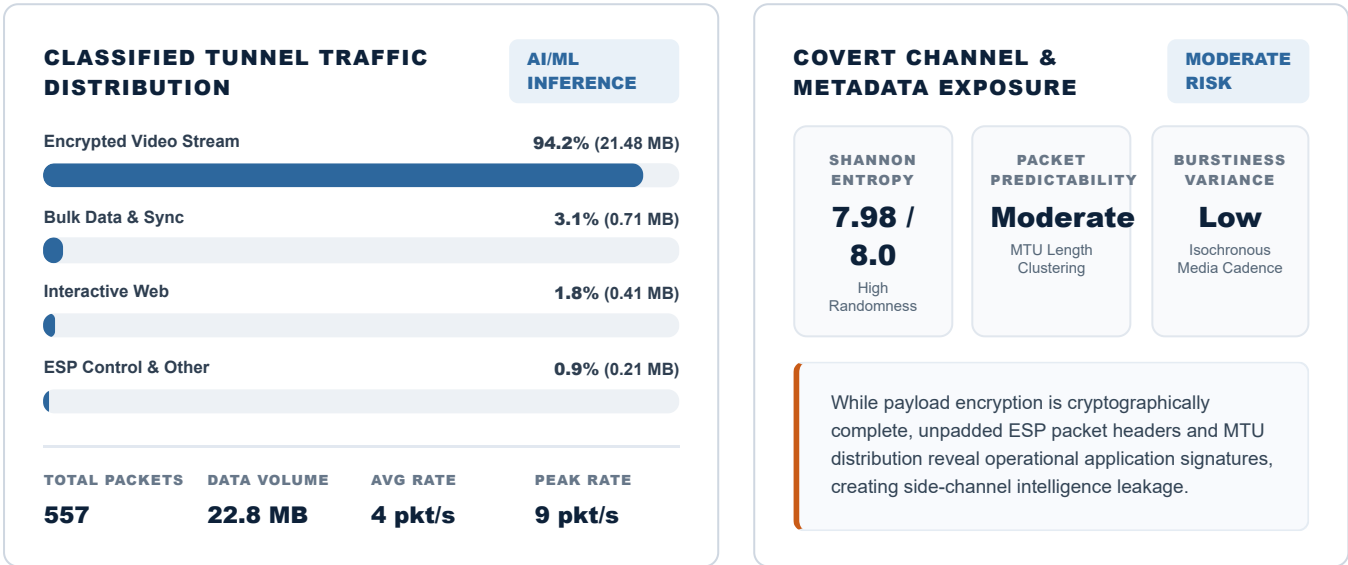
TRAFFIC CLASSIFICATION & STRATEGIC ROADMAP

# Traffic Overview & Strategic Remediation

Deep payload inference distribution, side-channel leakage, and prioritized resource allocation roadmap

## Section 4: High-Level Traffic Overview

Summary distribution of classified tunnel traffic and covert channel risk



## Section 5: Strategic Remediation Action Plan

Prioritized recommendations for leadership sign-off and operational resource allocation



## Executive Sign-Off & Platform Attestation



